MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION
FOURTH SEMESTER
Discipline Specific Elective (DSE) – MG4DSEMATA02 –
COMPUTATIONAL METHODS FOR OPTIMIZATION
(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 70

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Short Answer Questions

Answer any 5 questions
Each question carries 2 marks

1. Define an optimization problem and identify its basic elements. [K] / [CO1]
2. Write the general mathematical form of a constrained optimization problem. [K] / [CO1]
3. State the necessary condition for optimality in single-variable optimization. [K] / [CO2]
4. What is meant by Lagrange multipliers? [K] / [CO2]
5. Define a basic feasible solution in Linear Programming. [K] / [CO3]
6. What is meant by a unimodal function? [K] / [CO4]
7. Mention any two applications of Linear Programming. [K] / [CO3]
8. Write the golden-section ratio used in Golden Section search. [K] / [CO4]

[2x5 = 10]

Part B
Short Essay Questions

Answer any 5 questions
Each question carries 6 marks

9. Explain the classification of optimization problems based on constraints and design variables with examples. [U] / [CO1]

10. Formulate an optimization problem and classify it based on:

    (i) existence of constraints,

    (ii) nature of design variables.

    [A] / [CO1]

11. Solve the following using single-variable optimization: Find the stationary point and test whether it is a minimum/maximum for

$$f(x) = x^3 - 6x^2 + 9x + 1.$$

    [A] / [CO2]

12. Use the method of Lagrange multipliers to find the extremum of

$$f(x, y) = x^2 + y^2$$

    subject to

$$x + y = 10.$$

    [A] / [CO2]

13. Convert the following LP into standard form and identify initial BFS:

    Maximize

$$Z = 5x_1 + 3x_2$$

    subject to

$$2x_1 + x_2 \leq 8, \qquad x_1 + 2x_2 \leq 10, \qquad x_1, x_2 \geq 0.$$

    [A] / [CO3]

14. Explain the Two-Phase Simplex Method and its need. [U] / [CO3]

15. Describe the algorithm of the Golden Section Method for 1D optimization. [U] / [CO4]

16. Write notes on Steepest Descent Method and its convergence behaviour. [U] / [CO4]

[6x5 = 30]

Part C
Essay Questions

Answer any 3 questions
Each question carries 10 marks

17. a. Explain the statement of an optimization problem with design vector, constraints and objective function.

    b. Discuss classification based on physical structure of the problem.

[U] / [CO1]

18. Solve the following constrained optimization problem using Kuhn–Tucker conditions:

    Minimize

$$f(x, y) = x^2 + y^2$$

    subject to

$$x + y \geq 4, \qquad x \geq 0, \qquad y \geq 0.$$

    Clearly verify KKT conditions and obtain optimal solution. [An] / [CO2]

19. Solve the following LPP using Simplex Method:

    Maximize

$$Z = 3x_1 + 2x_2$$

    subject to

$$x_1 + x_2 \leq 4, \qquad 2x_1 + x_2 \leq 5, \qquad x_1, x_2 \geq 0.$$

    Find optimum solution and value of Z. [A] / [CO3]

20. a. Explain the Fibonacci Search Method.

    b. Compare Fibonacci and Golden Section methods (iterations, interval reduction, efficiency).

[U] / [CO4]

21. a. Derive the update formula of the Newton's method for multivariable optimization.

    b. Apply one iteration of Newton's method to minimize

$$f(x, y) = x^2 + xy + y^2 - 6x - 8y$$

    starting from $(0, 0)$.

[A] / [CO4]

22. Solve the following using Dual Simplex Method:

Minimize

$$Z = 2x_1 + 3x_2$$

subject to

$$-x_1 + x_2 \geq 1, \qquad x_1 + 2x_2 \geq 6, \qquad x_1, x_2 \geq 0.$$

Obtain optimal solution and interpret. [A] / [CO3]

[10x3 = 30]